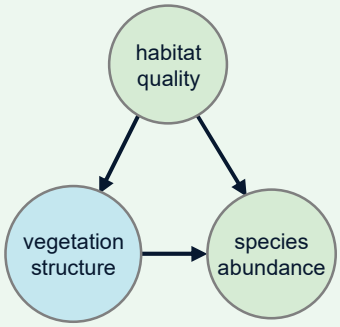
Scientific Evidence and Causal Inference

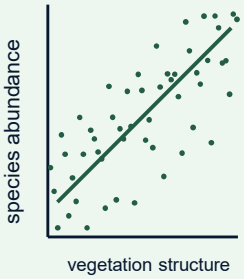
ZOO 4710-80 (Special Topics) • CRN: **25371**
Asynchronous Online • 3 Credits • J-Term II (January 4–29, 2027)
Instructor: Dr. Patrick Kelley (email: jkelle24@uwyo.edu)

For Zoology, Physiology, WFBM, Biology, and ENR majors

Prerequisites: LIFE-1010 + one 2000-level LIFE course (e.g. LIFE-2022 or LIFE-2023)
Enrollment also by permission of instructor.

Ecologists usually want to know what *causes* what. But are our statistical methods actually answering that question?





Why do wildlife populations decline? What shapes animal behavior and species distributions? These are **causal questions**, but complicated statistics do not necessarily provide causal answers. Through published studies and interactive investigations, students will challenge explanations, question underlying assumptions, and discover what evidence does and does not establish.

WHAT STUDENTS WILL EXPLORE

- How ecologists move from observations to explanations
- Why the data ecologists collect do not always support the conclusions they draw.
- How causal diagrams reveal hidden assumptions
- Why AIC-based multi-model inference does not, by itself, establish cause and effect
- How to evaluate interventions in ecology and conservation

COURSE FORMAT

- 10 self-paced instructional Modules (no scheduled meetings)
- Short video lectures, readings, interactive exercises
- Students may use their own research projects (but not required)
- Final week for students needing additional time to complete projects.

Build practical skills for evaluating research, identifying missing evidence, and drawing scientifically defensible conclusions about the natural world.